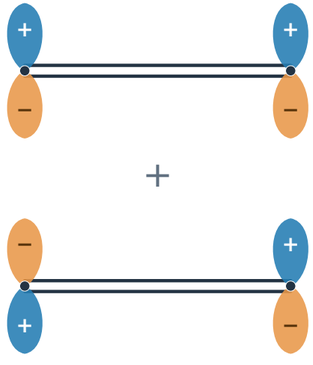
[2+2] cycloaddition | orbital phases along the reaction

THERMALLY ALLOWED (FORMAL)

suprafacial / antarafacial

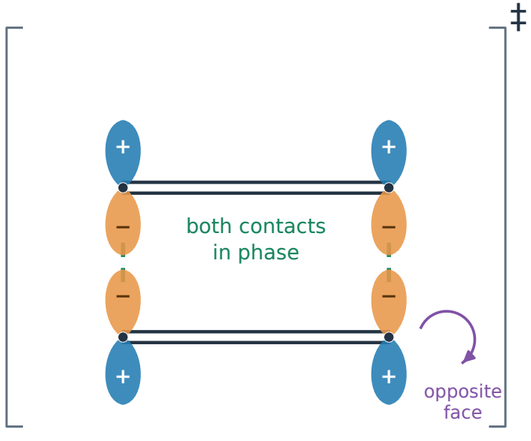
Möbius circuit · 4 electrons

Ethene HOMO, ψ₁ (occupied)



Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.

Δ

Product connectivity

Cyclobutane

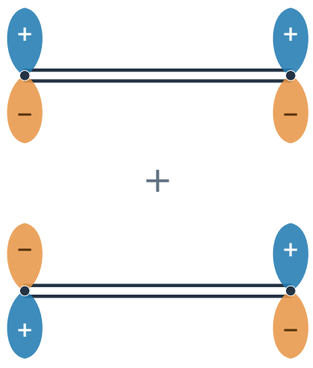
New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / suprafacial

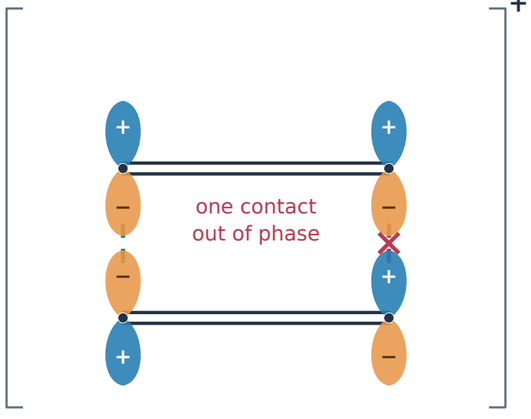
Hückel circuit · 4 electrons

Ethene HOMO, ψ₁ (occupied)



Ethene LUMO, ψ₂ (empty)

Δ



Partial bonds join the facing lobes.

Proposed product connectivity

Cyclobutane

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Antarafacial allowance is formal: geometry may prevent this approach, especially for ethene. Products show connectivity only.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.